

Plant A Maintenance Agent

An ontology-backed AI agent for field engineers in Japanese manufacturing

Every answer comes with evidence you can trace on a graph.
Gemini 3.8 Flash · Agent Development Kit · Vertex AI · Cloud Run · Neo4j AuraDB

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All data in this deck is synthetic ("Plant A"). No real company, product or person is referenced.

Maintenance know-how that lives in four systems and in retiring veterans can become an AI agent that answers with traceable evidence, on the data a plant already has



Problem

When a machine fails at night, the facts needed to decide are scattered across the equipment register, work orders, the parts inventory and manuals, and the veteran who connects them is off duty or retiring. Document search cannot connect them.



Solution

We model the plant as an ontology (equipment, components, failure modes, symptoms, work orders, procedures, parts, people) in a knowledge graph. A Gemini agent built with ADK walks it through read-only tools and cites work-order IDs and manual pages.



Proof

Deployed on Cloud Run with Neo4j AuraDB. In rehearsal, 45 of 45 reference answers were correct across diagnosis, cross-line comparison, parts and people, aggregation and “no record” questions. ADK hallucination score 1.0; median answer about 20 seconds.



Impact

Shorter diagnosis on every unplanned stop, cross-line patterns that registers alone hide, and know-how that survives retirements. The same pipeline runs on a plant's own registers, work orders and manuals, in Japanese or English, in four weeks.

Field engineers lose hours on every unplanned stop because the facts needed to decide live in four places, and in one veteran's head



Equipment register & BOM

- Models, install years and component structure in a CMMS or Excel
- Finding “the same model on another line” means searching across registers



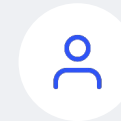
Work orders

- “What happened, when, what we did” in a CMMS, daily reports or paper
- Similar past cases are found from memory, if at all



Manuals

- Vendor PDFs with judgement thresholds and procedures
- Someone has to remember which page



Veteran experience

- Which cause to suspect from a symptom; what to check on a repeat failure
- Not on the night shift; lost at retirement

Plant A (our synthetic demo plant) Three press procedures (slide-guide, clutch-brake and flywheel-bearing replacement) can only be performed by one technician who holds the press-operation certificate and retires at the end of FY2026. The register, the work orders and the skills ledger each know one piece of this; nobody sees it until the three are connected.

Document search (RAG alone) cannot answer the questions that matter most on the floor: cross-line similarity, counts, and “who can do this”

Document search only

- Retrieves text chunks that look similar. The P-301 manual and P-301 reports are found
- Reports from the same model on another line never say “P-301”, so they are missed
- Cannot count: “which machines had the most bearing-related stops this year?”
- Evidence is a text fragment; a person must work out which machine and which report
- May infer a part number or lead time from context and state it as fact

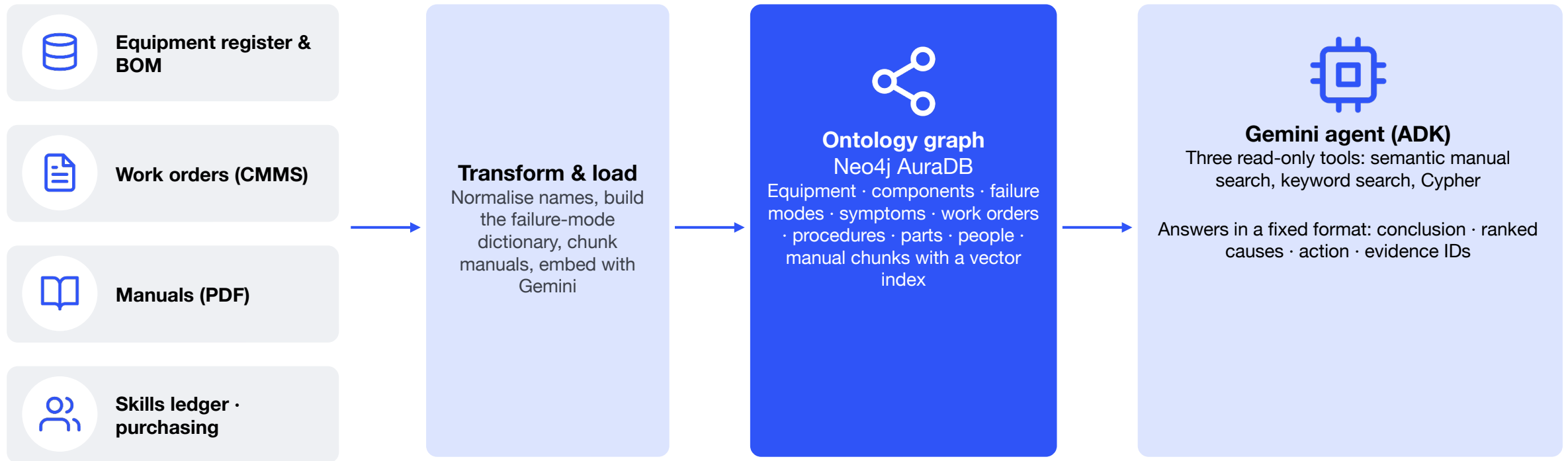


Knowledge graph + agent (ours)

- Stores equipment, components, failure modes and work orders as things and relationships
- Crosses lines through the model node: P-301 → CP-200 ← P-101
- Aggregates by failure-mode category and date in one query
- Evidence is work-order IDs, procedure IDs and manual pages, drawn as a sub-graph
- Answers “no record” when the graph has nothing; never invents an ID

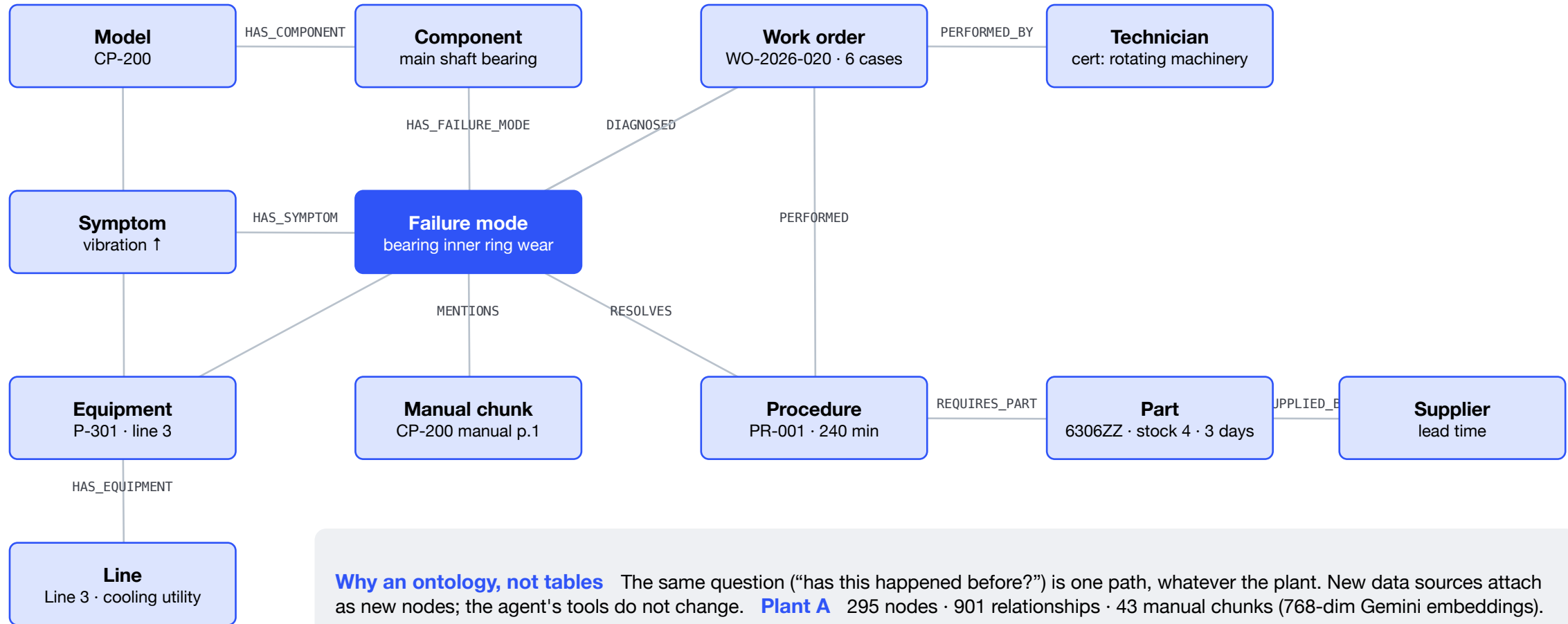
Vector search over the manuals finds where to start; the knowledge graph supplies the ranked answer, the next action and the evidence

A plant's existing data



Three mechanisms 1 Vector search picks the entry point (a manual passage linked to failure modes). 2 The graph walk delivers the answer: causes ranked by history, procedure, parts in stock, certified people. 3 Every ID in the answer is a node the user can inspect on screen.

Twelve node types and seventeen relationships, with the failure mode as the hub, cover diagnosis, history, the procedure, parts and people in one walk



3. The product

In the live app every cited ID is a clickable node: the engineer sees what the agent did, reads the answer, and checks the evidence on the graph

Plant A maintenance agent

Ask about a machine. Watch the agent search the manuals and walk the graph. Check the evidence yourself.

Example questions

Start over

Follow the structure

What model is P-301?

The cooling pump P-301 on line 3 is vibrating more than usual. What are the likely causes and how were they handled before?

Did pumps of the same model on other lines show the same failure?

For that repair, which parts are in stock, what are the lead times, and who is certified to do it?

Find what registers alone cannot show

Which machines use the 6306ZZ ball bearing? Are 4 in stock enough?

If Mr. Saeki (T-01) retires, who can take over his work?

Which procedures need a part with zero stock, and how often were they performed?

Count and compare

Which machines had the most bearing-related stops in the last 12 months?

Total downtime per line, and the most frequent failure mode on each?

Search the manuals

At what vibration level does a pump need attention?

Which machine and component uses bearing NU320?

Say no when there is no record

Were there any bearing-related failures on P-302?

What is the status of P-401?

The cooling pump P-301 on line 3 is vibrating more than usual. What are the likely causes and how were they handled before?

What the agent did

Answered in 38.2s

Searched the manuals "vibration increase abnormal vibration" 3 passages · 5.6s

Queried the knowledge graph 4 rows · 4.9s + Cypher

Conclusion

Vibration increase on cooling pump P-301 is most frequently caused by main shaft bearing inner ring wear, followed by coupling misalignment and impeller imbalance. In previous events, issues were addressed primarily by bearing replacement, coupling realignment, or impeller cleaning.

Candidate causes

Main shaft bearing inner ring wear (主軸ベアリング内輪摩耗): past cases: 6

Coupling misalignment (カップリング芯ずれ): past cases: 2

Impeller imbalance (foreign matter buildup/wear) (インペラ不釣り合い (異物付着・摩耗)): past cases: 2

Bearing grease degradation (軸受グリース劣化): past cases: 2

Recommended action

Conduct vibration analysis and perform Cooling Pump Main Shaft Bearing Replacement (冷却ポンプ主軸ベアリング交換, PR-001), which requires Rotating Machinery Maintenance (回転機械整備) certification.

Evidence

Work orders: WO-2024-006, WO-2025-008, WO-2025-015, WO-2025-028, WO-2025-030, WO-2025-037, WO-2026-002, WO-2026-015, WO-2026-020, WO-2026-026

Procedure: PR-001, PR-003, PR-004

Manual: CH-cp-200-002 (CP-200 渦巻型冷却水ポンプ 保全マニュアル 第3版, p.1)

Path: P-301 → CP-200 → 主軸ベアリング → 主軸ベアリング内輪摩耗 → WO-2026-020 → PR-001

Ask about a machine, a failure, a part or a technician...

Ask

How it works

Source

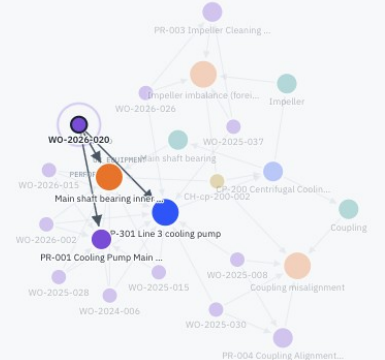
EN

日本語

Evidence

The cooling pump P-301 on line 3 is vibrating more than usual. What are the likely causes and how were they handled before?

22 nodes · 44 edges



Lines, equipment, models

Components, parts, suppliers

Failure modes, symptoms

Work orders, procedures, technicians

Manual pages

1 What the agent did

Semantic manual search, then one Cypher query. Each step shows the search text or the Cypher, row counts and seconds.

2 Answer in a fixed format

Conclusion · candidate causes with past-case counts · recommended action with the required certification · evidence IDs.

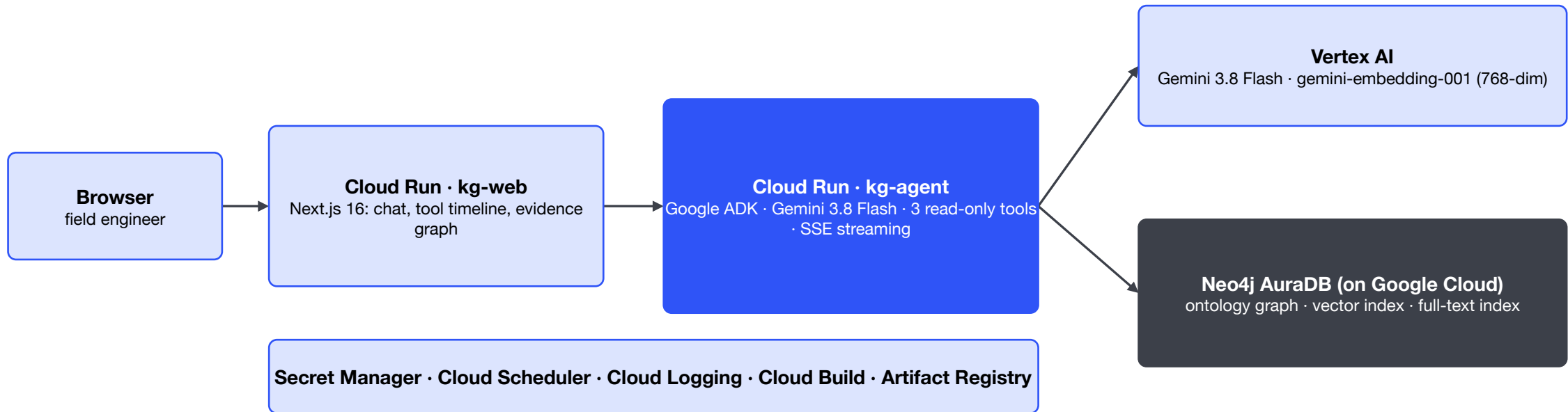
3 Evidence graph

Shortest paths between the cited nodes. Clicking WO-2026-020 in the answer highlights it on the graph.

4 Bilingual over Japanese records

Ask in English or Japanese; the graph values stay in Japanese with English names, IDs never change.

An ADK agent with three read-only tools, Gemini for reasoning and embeddings, two Cloud Run services, and guardrails that reject writes and invented IDs



Read-only by construction

Write clauses are rejected before execution; transactions open in READ mode; 20-second query timeout. Covered by unit tests.

Evidence is mandatory

Fixed answer format with an Evidence section. IDs become chips and are resolved against the graph, so an unknown ID cannot appear as a node.

Private agent, public UI

kg-agent accepts only ID tokens from the web service account and Cloud Scheduler. Neo4j credentials come from Secret Manager.

Evaluated, not assumed

ADK eval set with LLM-judged response match and hallucination metrics, plus a rehearsal harness with fact checkers (next slide).

Rehearsal and ADK evaluation: 45 of 45 reference answers correct, hallucination score 1.0, median answer about 20 seconds with two to three tool calls

45 / 45

reference answers correct
(9 cases × 5 runs, fact-checked)

1.0

ADK hallucinations_v1
(7 / 7 eval cases pass)

≈ 20 s

median answer time
2–3 tool calls

0

invented work-order IDs,
parts or people

Case	Question	Correct	Median (min–max)	Tool calls
Q1	P-301 vibrating: causes and past fixes (JA / EN)	10 / 10	23 s (18–32)	2.4
Q2	Same model on other lines (follow-up)	5 / 5	32 s (22–42)	2.0
Q3	Parts in stock, lead times, certified technicians	5 / 5	24 s (22–30)	2.2
Q4	Most bearing-related stops in the last 12 months	5 / 5	26 s (20–37)	2.6
B2	Who can take over when T-01 retires	5 / 5	35 s (19–44)	3.2
D2	Which machine uses bearing NU320 (keyword)	5 / 5	20 s (18–44)	3.0
E1 / E2	Bearing failures on P-302 · unknown machine → “no record”	10 / 10	19–23 s	2.4–3.4

Rehearsal on the deployed AuraDB graph with gemini-3.8-flash, 2026-10-06 (agent/rehearse.py, log in docs/eval). ADK eval: final_response_match_v2 = 1.00 on all 7 cases, hallucinations_v1 = 1.00 on 6 cases and 0.86 on the three-turn conversation (judge gemini-3.5-flash, 3 samples).

For a plant the value is hours of downtime avoided on every incident, patterns that registers hide, and know-how that survives retirements

80

unplanned stops in two years at Plant A

935 min

bearing-related downtime on one pump in 12 months

1

technician holding the only press certification, retiring FY2026

≈ 20 s

from question to ranked causes, parts and people



Faster diagnosis on every stop

- Plant A: 80 unplanned stops in two years, 9 bearing-related in the last 12 months alone (P-301: 3 stops, 935 minutes).
- The agent returns ranked causes, the procedure, parts and people in about 20 seconds instead of a search across four systems.
- Illustrative: one hour saved per incident × 40 incidents a year × ¥500k per hour of line stop (assumption) ≈ ¥20M a year per plant.



Patterns registers hide

- Same-model failures across lines (P-101 ×4, P-201 ×1) surface in one answer.
- A part shared by 8 machines with stock for two replacements; procedures that depend on a zero-stock part.
- Bearing-related downtime concentrated on the CP-200 pumps: a maintenance-plan decision, not a repair.



Know-how that survives

- One technician holds the only press certification and retires in FY2026; the graph shows which procedures have no successor.
- Work orders become searchable history; manual pages become cited evidence.
- New engineers and night shifts get the veteran's path, with the IDs to verify it.

Cost figures are illustrative assumptions for a mid-size plant; the team's manufacturing CFO is validating them against industry benchmarks for the final pitch.

What is new: answers grounded in an ontology with graph evidence, bilingual access to Japanese floor documents, and an agent that is read-only and says “no record”



Ontology-grounded GraphRAG

- Vector search only chooses where to start; the content of the answer comes from graph structure, so every claim maps to a node and a path.
- Cross-line, aggregation and “who can do it” questions that document RAG cannot answer become one Cypher query.



Built for Japanese plants

- Registers, work orders and manuals stay in Japanese; the master data carries Gemini-generated English names.
- An engineer asks in either language and gets the same IDs. JAPAC plants with mixed teams get one assistant.



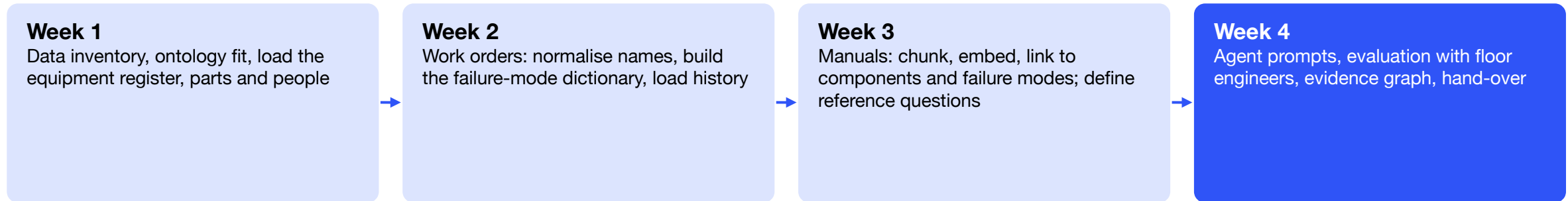
Honest and safe by design

- The agent cannot write to the graph and rejects write clauses before execution.
- When there is no record it says so and lists only the IDs it checked. Verified in rehearsal and ADK eval.

Why Gemini + ADK made this possible in days

Gemini 3.8 Flash writes correct Cypher against a schema it is shown once and keeps a fixed answer format across languages; gemini-embedding-001 indexes Japanese manuals without a local model; ADK gives the tool loop, SSE streaming to the UI, Cloud Run deployment and an evaluation harness. The whole build, from synthetic plant to deployed app, took one week.

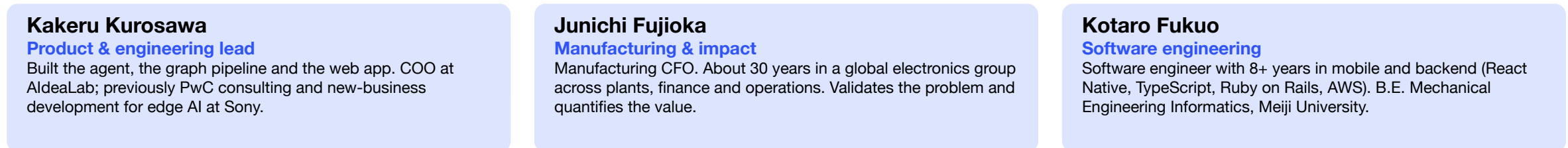
Four weeks from a plant's own registers to a working assistant; next: sensor time series, approved write-back, and a mobile UI for the floor



After the first four weeks



Team



Try it, read the code, ask it a question.

Live app kg-web-7ikzkb2evq-an.a.run.app

Source github.com/kurorooooo/kg-maintenance-agent

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